

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1023	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	N. Lahari	Reg. No.:	192473018
Date:	09-10-2026	Duration:	60 minutes

Code of Conduct

I n . l a h a r i (1 9 2 4 7 3 0 1 8) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N. Lahari_____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Agile Sprint Planning & User Story Workshop

1. Project Vision and Stakeholder Identification

1.1 Project Vision

The main aim of this project is to develop an AI-assisted customer support system that automatically answers common customer questions and transfers complex, sensitive, or uncertain queries to human support agents.

The system uses artificial intelligence to understand customer queries, identify the type of request, and provide suitable responses. When the AI cannot confidently solve a problem, the conversation is transferred to a human agent.

The project aims to reduce customer waiting time, improve support quality, maintain data security, and provide reliable customer service.

1.2 Key Stakeholders

Stakeholder	Role and Responsibility
Customers	Ask questions and receive support.
Support Agents	Handle complex and escalated queries.
Product Owner	Defines requirements and priorities.
Development Team	Designs and develops the system.
Testing Team	Checks system functionality and quality.
Operations/IT Team	Maintains infrastructure and security.
Management	Monitors performance and business benefits.

1.3 System Workflow

The system follows these steps:

1. The customer submits a query.
2. The AI chatbot analyses the query.
3. If the query is routine, the chatbot provides an automatic response.
4. If the query is sensitive, complex, or uncertain, it is transferred to a human agent.

5. The agent resolves the issue and updates the conversation record.

2. Epics and User Stories

An epic is a large feature of the project, while a user story describes a smaller feature from the user's point of view.

User stories follow this format:

“As a user, I want a feature, so that I can receive a benefit.”

2.1 Epics

Epic ID	Epic Name	Description
E1	AI Chatbot	Answers common customer questions automatically.
E2	Human Escalation	Transfers complex queries to human agents.
E3	Secure Data Management	Stores conversations and maintains audit records.
E4	API Integration	Connects the system with external services.
E5	Monitoring and Analytics	Tracks response quality and system performance.
E6	Responsible AI	Supports fairness, privacy, and transparency.

2.2 User Stories

US1 – Automatic Customer Support

As a customer, I want the chatbot to answer common questions, so that I can receive quick support.

US2 – Human Escalation

As a customer, I want my query to be transferred to a human agent when the AI cannot solve it, so that I can receive proper assistance.

US3 – Secure Conversation Records

As a support manager, I want conversations to be stored securely, so that I can review previous customer interactions.

US4 – External API Integration

As a system administrator, I want the platform to connect with external services, so that customer

information can be retrieved when required.

US5 – Performance Monitoring

As a product owner, I want to monitor response time and chatbot accuracy, so that I can improve system performance.

US6 – Privacy and Fairness

As a customer, I want my personal information to be protected and responses to be fair, so that I can trust the service.

3. Acceptance Criteria

Acceptance criteria define the conditions that must be satisfied before a user story is accepted.

The following criteria use the Given–When–Then format.

US1: Automatic Customer Support

- Given a customer submits a common question, when the chatbot identifies the question, then it should provide a relevant response.
- Given the chatbot cannot understand the query, when confidence is below the configured threshold, then it should ask for clarification or offer human assistance.

US2: Human Escalation

- Given a customer submits a sensitive or complex query, when the system identifies the need for human assistance, then it should transfer the conversation to a support agent.
- Given a conversation is transferred, when the agent opens it, then the available conversation history should be displayed.

US3: Secure Conversation Records

- Given a conversation is completed, when the system saves the interaction, then it should record the relevant details securely.
- Given an unauthorised user attempts to access a record, when the system checks permissions, then access should be denied.

US4: API Integration

- Given an authorised external service is available, when the system sends a valid request, then it should process the response correctly.
- Given the external service fails, when the request cannot be completed, then the system should handle the error safely.

US5: Performance Monitoring

- Given the chatbot is operating, when interactions are processed, then the system should collect relevant performance metrics.
- Given a configured performance threshold is exceeded, when monitoring detects the issue, then an alert should be generated.

US6: Privacy and Fairness

- Given a customer shares personal information, when the system processes the query, then it should follow applicable privacy and access-control rules.
- Given the AI generates a response, when the response is checked, then it should follow the system's safety and fairness requirements.

4. Product Backlog Prioritization

The product backlog contains all features and tasks required to develop the project. The MoSCoW method is used to prioritise them.

MoSCoW stands for Must Have, Should Have, Could Have, and Won't Have for now.

Priority	Backlog Item	Reason
Must Have	AI chatbot for routine queries	Provides the main customer support function.
Must Have	Human escalation	Handles complex and sensitive cases.
Must Have	Secure conversation records	Protects customer information and supports auditing.
Must Have	Basic error handling	Prevents failures from disrupting the service.
Should Have	External API integration	Improves access to customer information.
Should Have	Monitoring and alerts	Helps identify system problems.
Should Have	Basic analytics dashboard	Helps evaluate performance.
Could Have	Multi-language support	Makes the system accessible to more users.
Could Have	Advanced analytics	Provides deeper performance insights.
Won't Have for Now	Voice-based support	Can be considered in a future release.

4.1 Prioritization Diagram

Figure 2: MoSCoW Product Backlog Prioritization



The Must Have features will be developed first because they are essential to the project's basic operation. Should Have features will be planned next, while Could Have features may be added if time permits.

5. Sprint Backlog and Actionable Tasks

A sprint backlog contains the selected user stories and the tasks required to complete them during a sprint.

For this project, a two-week sprint is proposed to develop the initial working version of the customer support system.

5.1 Sprint Backlog

Task ID	Task Description	Estimated Story Points
T1	Set up the project and chatbot model	5
T2	Implement query classification	5
T3	Develop human escalation logic	8
T4	Integrate required APIs	8
T5	Build secure conversation logging	5
T6	Create a basic monitoring dashboard	5
T7	Perform unit and integration testing	3

Task ID	Task Description	Estimated Story Points
T8	Prepare deployment and configuration	3
Total		42

5.2 Sprint Activities

Week 1: Planning and Development

- Finalise requirements and select user stories.
- Set up the development environment.
- Develop the chatbot and query classification.
- Implement human escalation logic.
- Begin secure conversation logging.

Week 2: Integration and Testing

- Integrate the required APIs.
- Develop the basic monitoring dashboard.
- Test chatbot responses and escalation behaviour.
- Fix identified defects.
- Demonstrate the working system during the sprint review.

The team will conduct a short daily Scrum meeting to discuss completed work, planned work, and any obstacles.

6. Story Point Estimation

Story points are used to estimate the relative effort, complexity, and uncertainty of a user story. The Fibonacci sequence is commonly used for estimation.

The sequence used in this project is 1, 2, 3, 5, 8, and 13.

6.1 Estimation Table

User Story	Story Points	Justification
US1 – Automatic Customer Support	5	Requires chatbot setup and response generation.

User Story	Story Points	Justification
US2 – Human Escalation	8	Requires classification, routing, and conversation transfer.
US3 – Secure Conversation Records	5	Requires secure storage and access control.
US4 – External API Integration	8	Requires integration and error handling.
US5 – Performance Monitoring	5	Requires metrics collection and dashboard development.
US6 – Privacy and Fairness	5	Requires privacy rules and response checks.
Total Estimated Points	36	

The story point estimates represent relative complexity, not the number of hours needed. The development team will review and adjust the estimates as it learns more about the requirements.

The sprint backlog estimates and user story estimates are separate planning views. The team should ensure that the selected stories and tasks match before finalising the sprint commitment.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

The goal of the sprint is to develop a working AI-assisted customer support prototype that answers routine questions, transfers complex or sensitive cases to human agents, stores conversation records securely, and provides basic performance monitoring.

7.2 Expected Deliverables

At the end of the sprint, the team expects to deliver:

1. A functional AI chatbot for common customer questions.
2. A human escalation mechanism for complex or uncertain queries.
3. Secure conversation logging with appropriate access controls.
4. Initial API integration and error handling.
5. A basic monitoring dashboard.
6. Unit and integration test results.

7. A working prototype for demonstration.
8. Project documentation and sprint review notes.

7.3 Definition of Done

A user story will be considered complete when:

- The required functionality has been implemented.
- The acceptance criteria have been satisfied.
- Relevant tests have passed.
- Identified critical defects have been resolved.
- The code has been reviewed and integrated.
- The feature is demonstrated to the product owner.

7.4 Agile Sprint Cycle

The team follows an iterative development process. Each sprint begins with planning, continues through development and testing, and ends with a review and retrospective.

Figure 3: Agile Sprint Cycle



The sprint review collects feedback on the working product. The retrospective identifies improvements for the next sprint.

Conclusion

The AI-Assisted Customer Support with Human Escalation Control project demonstrates how Agile software development can be applied to an intelligent customer service system. Project vision, stakeholders, epics, user stories, acceptance criteria, backlog prioritisation, sprint planning, and story point estimation provide a structured development approach.

The system aims to answer routine questions quickly while ensuring that complex or sensitive cases receive human attention. Secure data management, monitoring, privacy, and fairness are also important considerations. By following short sprints and collecting regular feedback, the development team can improve the system incrementally and deliver a useful customer support solution.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification (30%)	/5
Epics & User Stories (25%)	/5
Acceptance Criteria (20%)	/5
Product Backlog Prioritization & Sprint Planning (15%)	/5
Story Point Estimation, Sprint Goal & Presentation (10%)	/5

TOTAL	/25
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